Chapter 5 — States and Phases of Matter

Complete Notes · PECTAA Chemistry 11 · Board-style longs + conceptual shorts

Topic 1 — Gases, ideal gas equation and kinetic theory

Q.1 Describe the properties of gases, derive the ideal gas equation and explain the kinetic molecular theory.

Gases have no fixed shape or volume, are highly compressible, diffuse rapidly and exert pressure. According to kinetic molecular theory, gas particles are in constant random motion, their volume is negligible compared with the container, intermolecular forces are negligible in an ideal gas, and average kinetic energy depends on absolute temperature.

Boyle's law gives $V \propto 1/P$ at constant n and T; Charles' law gives $V \propto T$ at constant n and P; Avogadro's law gives $V \propto n$ at constant P and T. Combining these gives $PV = nRT$. With P in atm, V in dm^3 and T in K, $R = 0.0821\ atm\ dm^3\ K^{-1}\ mol^{-1}$. The equation can also be used to find molar mass: $M = mRT/PV$.

Important Conceptual Short Questions — Topic 1

1. Why are gases much more compressible than liquids?

Gas particles are far apart and most of the volume is empty space. Applying pressure can push them closer together, while liquid particles are already close.

2. How does kinetic theory explain gas pressure?

Gas particles collide with the walls of the container. The force of these collisions per unit area is observed as pressure.

3. Why must temperature be in kelvin in gas-law calculations?

Gas volume and pressure are proportional to absolute temperature. The kelvin scale starts at absolute zero, so ratios such as V/T are meaningful.

4. How does increasing temperature increase the pressure of a fixed volume of gas?

Higher temperature increases the average kinetic energy of particles. They collide with the walls more frequently and more forcefully, increasing pressure.

Topic 2 — Intermolecular forces and hydrogen bonding in water

Q.2 Explain the types of intermolecular forces and the influence of hydrogen bonding on the properties of water.

The main intermolecular forces are London dispersion forces, permanent dipole-dipole forces and hydrogen bonding. London forces exist in all molecules and increase with size and polarizability. Dipole-dipole forces occur between polar molecules. Hydrogen bonding occurs when H is bonded to N, O or F and is attracted to a lone pair on another molecule.

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Hydrogen bonding explains water's unusually high boiling point, high surface tension, high specific heat, high heat of vaporization and low vapour pressure. In ice, hydrogen bonds form an open lattice, making ice less dense than liquid water. This is why ice floats and why water expands on freezing.

Important Conceptual Short Questions — Topic 2

1. Why does water have a higher boiling point than methane?

Water molecules form hydrogen bonds, while methane molecules have only weak London forces. Much more energy is needed to separate water molecules.

2. How does hydrogen bonding make ice less dense than liquid water?

Hydrogen bonds hold water molecules in an open, ordered lattice with empty spaces. Liquid water has a more compact arrangement, so ice is less dense.

3. Why does HF show stronger hydrogen bonding than HCl?

Fluorine is much more electronegative and smaller than chlorine, so the H–F bond is highly polar and HF molecules form strong hydrogen bonds. HCl does not form comparable hydrogen bonds.

4. How do London forces explain the physical states of halogens at room temperature?

Down the group, halogen molecules become larger and more polarizable, so London forces increase. F_2 and Cl_2 are gases, Br_2 is a liquid and I_2 is a solid.

Topic 3 — Surface tension, viscosity, evaporation and boiling point

Q.3 Explain surface tension, viscosity, evaporation, vapour pressure and boiling point, and the factors affecting them.

Surface tension is the inward pull on surface molecules caused by unbalanced intermolecular forces. Viscosity is resistance to flow; it increases with stronger intermolecular forces and molecular size, and decreases with temperature. Evaporation occurs at the surface when high-energy molecules escape; it causes cooling.

Vapour pressure is the pressure of vapour in equilibrium with its liquid. It increases with temperature and decreases with stronger intermolecular forces. Boiling occurs when vapour pressure equals external pressure. Therefore boiling point rises with external pressure and falls at high altitude. Stronger intermolecular forces give higher boiling points.

Important Conceptual Short Questions — Topic 3

1. Why does evaporation produce cooling?

The fastest molecules escape from the liquid surface. The average kinetic energy of the remaining molecules decreases, so the temperature falls.

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2. Why does water have a higher surface tension than ethanol?

Water forms a stronger and more extensive hydrogen-bonded network. The inward pull on surface molecules is therefore greater.

3. How does temperature affect viscosity and vapour pressure?

Increasing temperature lowers viscosity because molecules move more easily past one another. It raises vapour pressure because more molecules have enough energy to escape.

4. Why does a liquid boil at a lower temperature on a mountain?

Atmospheric pressure is lower at high altitude. The liquid's vapour pressure equals this lower external pressure at a lower temperature.

Topic 4 — Energetics of phase changes

Q.4 Explain molar heat of fusion and vaporization and their importance in the study of ice sheets.

Molar heat of fusion is the heat required to melt one mole of a solid at its melting point. Molar heat of vaporization is the heat required to vaporize one mole of a liquid at its boiling point. These energies overcome intermolecular forces without changing temperature during the phase change.

Water has a high heat of fusion and a very high heat of vaporization because of hydrogen bonding. This means ice sheets and glaciers absorb large amounts of heat before melting, and melting ice moderates temperature changes in polar regions. The high heat of vaporization also makes sweating an effective cooling process.

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Important Conceptual Short Questions — Topic 4

1. Why does temperature remain constant during melting?

The supplied heat is used to overcome intermolecular forces and change the arrangement of particles, not to increase their average kinetic energy.

2. How does the high heat of fusion of ice affect polar ice caps?

A large amount of heat is needed to melt ice, so ice sheets resist rapid melting and help stabilize polar temperatures.

3. Why is heat of vaporization usually greater than heat of fusion?

Vaporization separates particles almost completely, while fusion only loosens the solid lattice. More energy is needed to overcome intermolecular forces fully.

4. How does sweating cool the body?

Evaporation of sweat absorbs heat from the skin because water has a high heat of vaporization, lowering body temperature.

Topic 5 — Solids, crystalline and amorphous forms, and liquid crystals

Q.5 Differentiate crystalline and amorphous solids and describe the properties and uses of liquid crystals.

Crystalline solids have a regular three-dimensional arrangement of particles, sharp melting points, definite cleavage planes and anisotropic properties. Examples include NaCl, diamond and ice. Amorphous solids lack long-range order, melt over a range of temperature and are isotropic. Examples include glass, rubber and plastic sulphur.

Liquid crystals have properties between liquids and crystals. They flow like liquids but have some ordered arrangement and optical properties like crystals. They are used in LCD displays, temperature sensors and diagnostic devices because their optical behaviour changes with temperature or electric field.

Property	Crystalline solid	Amorphous solid
Arrangement	Regular, long-range order	No long-range order
Melting	Sharp melting point	Melts over a range
Directional properties	Anisotropic	Isotropic
Example	NaCl, diamond	Glass, rubber

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Important Conceptual Short Questions — Topic 5

1. Why do crystalline solids have sharp melting points?

Their particles are arranged in a regular lattice with uniform forces. A definite amount of energy is needed to overcome these forces at one temperature.

2. How do amorphous solids differ from crystalline solids in melting behaviour?

Amorphous solids have regions with different local arrangements and bond strengths, so they soften and melt over a range rather than at a sharp temperature.

3. Why are liquid crystals useful in displays?

Their molecular orientation can be changed by an electric field, which changes how they transmit or rotate light. This allows pixels to be switched on and off.

4. How can liquid crystals be used in temperature sensing?

Some liquid crystals change colour or optical properties with temperature because their molecular ordering changes. This makes them useful in thermal strips and diagnostic devices.